

Graphs in Machine Learning

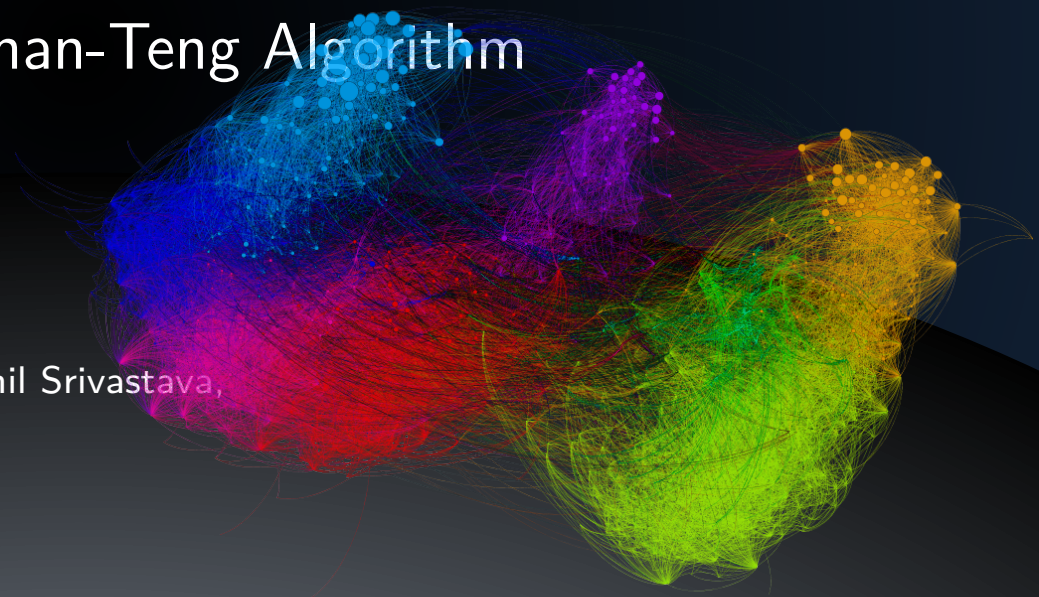
Spectral Graph Sparsifiers: Theory

Effective Resistance and Spielman-Teng Algorithm

Michal Valko

Isara Labs

Partially based on material by: Rob Fergus, Nikhil Srivastava, Yiannis Koutis, Joshua Batson, Daniel Spielman



Spectral Graph Sparsifiers

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As a consequence, $\arg\min_{\mathbf{x}} \|\mathbf{L}_H \mathbf{x} - \mathbf{b}\| \approx \arg\min_{\mathbf{x}} \|\mathbf{L}_G \mathbf{x} - \mathbf{b}\|$

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- a single pass over the data

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Laplacian smoothing (denoising): given $\mathbf{y} \triangleq \mathbf{f}^\star + \xi$ and G compute

$$\min_{\mathbf{f} \in \Re^n} (\mathbf{f} - \mathbf{y})^\top (\mathbf{f} - \mathbf{y}) + \lambda \mathbf{f}^\top \mathbf{L}_G \mathbf{f} \tag{1}$$

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Need to approximate spectrum only up to regularization level λ

Ridge Spectral Graph Sparsifiers in ML

Definition

An (ε, γ)-sparsifier of G is a reweighted subgraph H s.t.

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RLA $\rightarrow$ Graph: Improve over $\mathcal{O}(n \log n)$ exploiting regularization

Graph $\rightarrow$ RLA: Exploit $\mathbf{L}_G$ structure for fast (ε, γ)-sparsification

Spectral Graph Sparsification: Intuition

Let us consider unweighted graphs: $w_{ij} \in \{0, 1\}$

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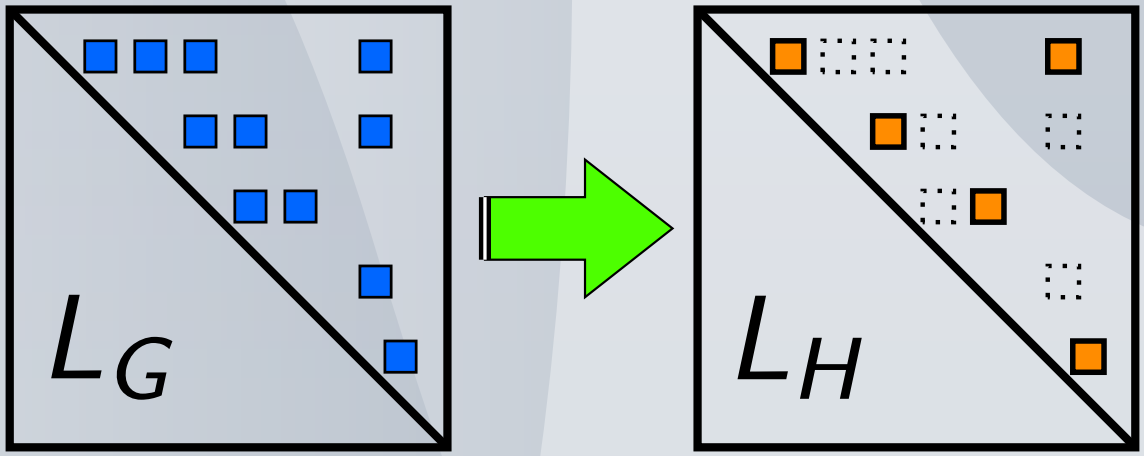
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https://math.berkeley.edu/~nikhil/

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Then $\sum_{e \in E} s_e \mathbf{v}_e \mathbf{v}_e^\mathsf{T} \approx \mathbf{I} \iff \sum_{e \in E} s_e \mathbf{a}_e \mathbf{a}_e^\mathsf{T} \approx \mathbf{A}$

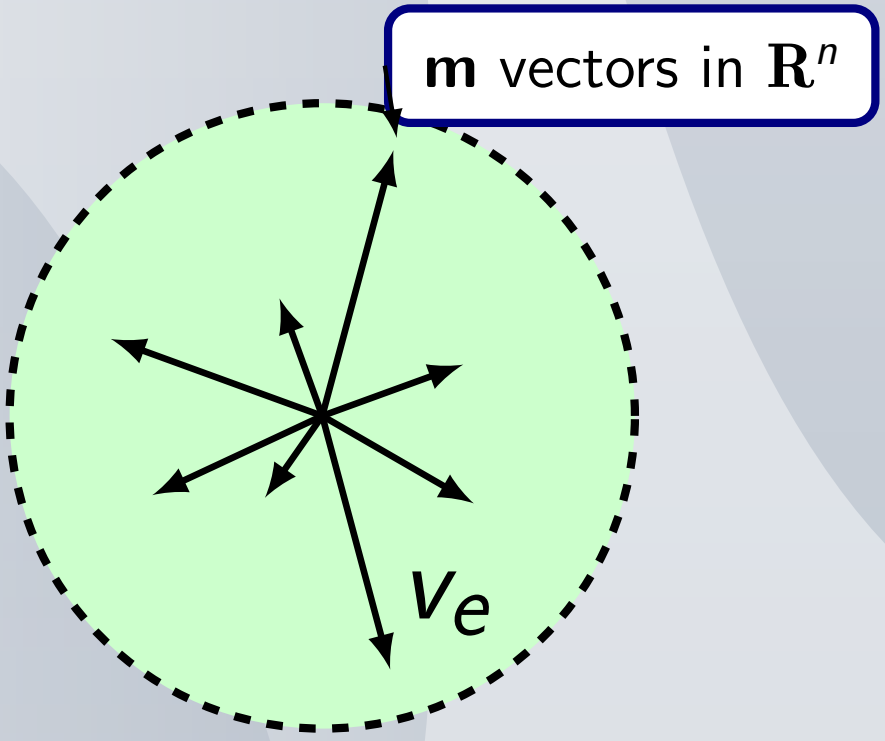
multiplying by $\mathbf{A}^{1/2}$ on both sides

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How does $\sum_{e \in E} \mathbf{v}_e \mathbf{v}_e^\top = \mathbf{I}$ look like geometrically?

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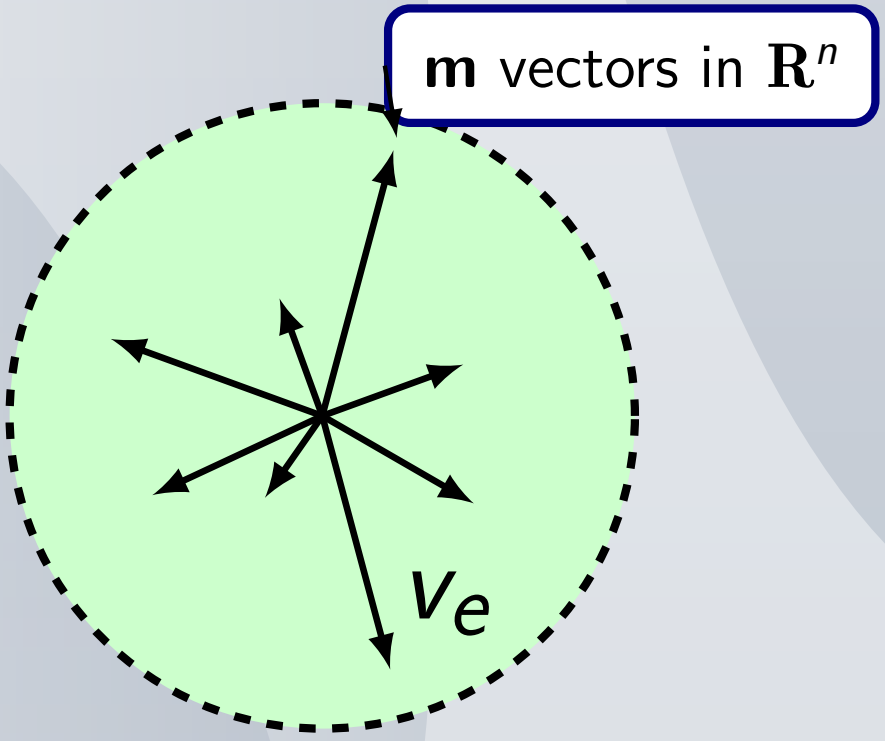
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moment ellipse is a sphere

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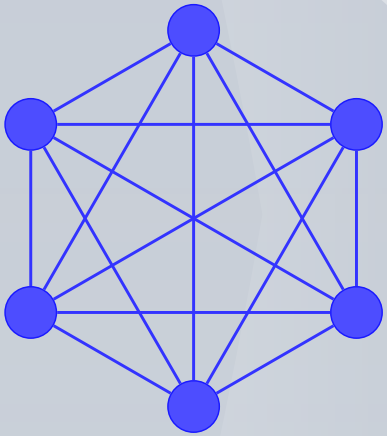
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Example: What happens with K_n?

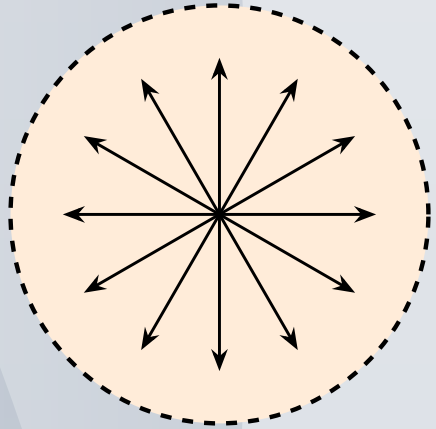
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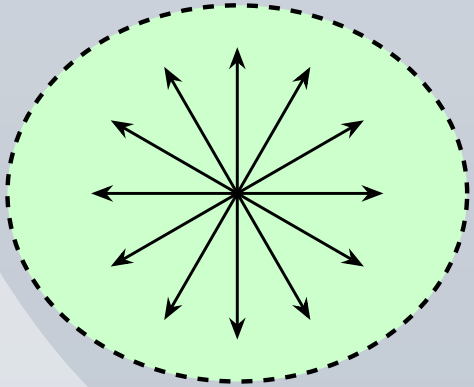
K_n graph



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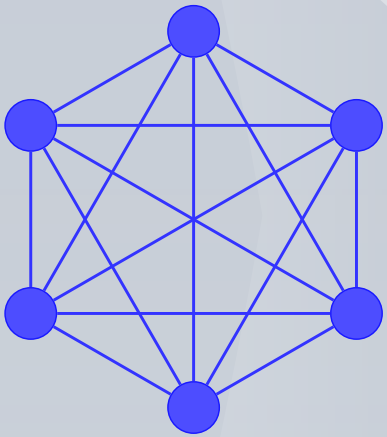
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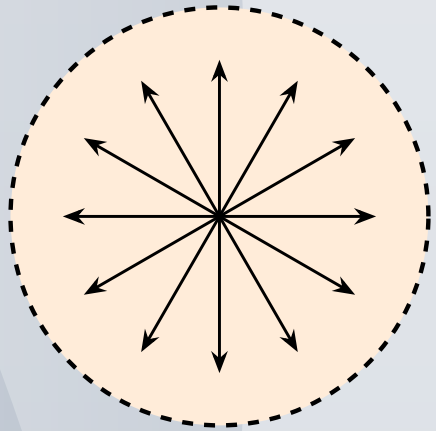
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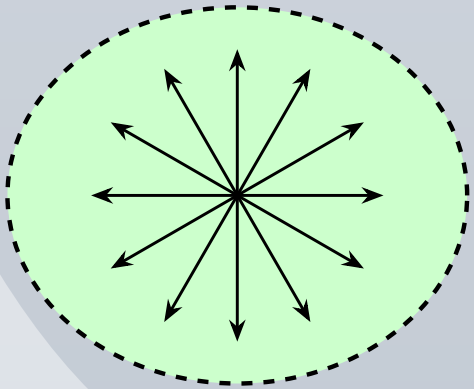
K_n graph



$\sum_{e \in E} \mathbf{b}_e \mathbf{b}_e^\top = \mathbf{L}_G$



$\sum_{e \in E} \mathbf{v}_e \mathbf{v}_e^\top = \mathbf{I}$

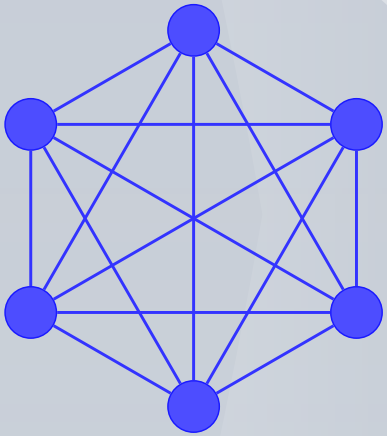


It is already isotropic! (looks like a sphere)

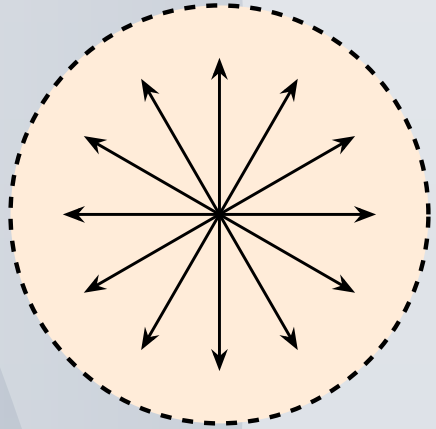
Spectral Graph Sparsification: Intuition

Example: What happens with K_n?

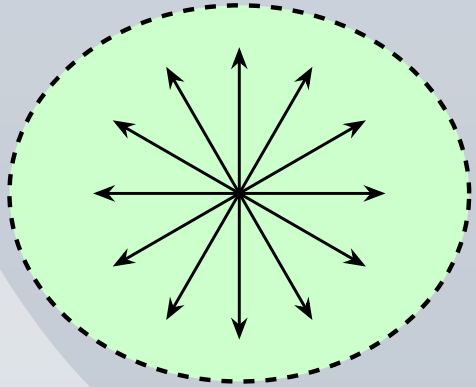
K_n graph



$\sum_{e \in E} \mathbf{b}_e \mathbf{b}_e^\top = \mathbf{L}_G$



$\sum_{e \in E} \mathbf{v}_e \mathbf{v}_e^\top = \mathbf{I}$



It is already isotropic! (looks like a sphere)

rescaling $\mathbf{v}_e = \mathbf{L}^{-1/2} \mathbf{b}_e$ does not change the shape

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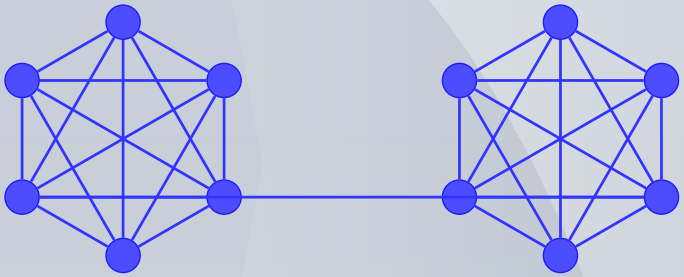
Spectral Graph Sparsification: Intuition

Example: What happens with a dumbbell?

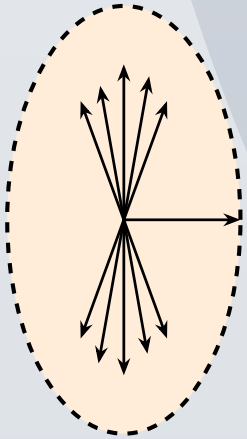
Spectral Graph Sparsification: Intuition

Example: What happens with a dumbbell?

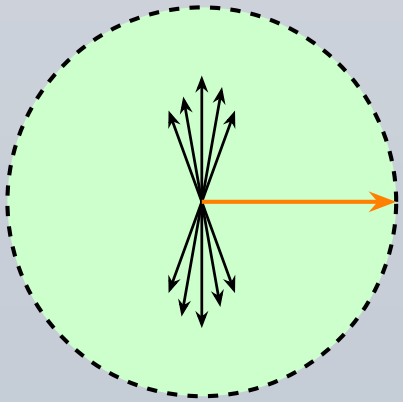
Dumbbell



$\sum_{e \in E} \mathbf{b}_e \mathbf{b}_e^\top = \mathbf{L}_G$



$\sum_{e \in E} \mathbf{v}_e \mathbf{v}_e^\top = \mathbf{I}$



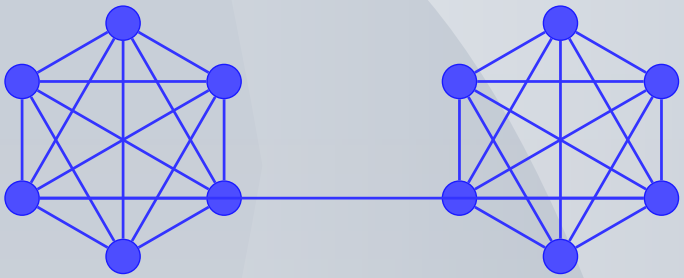
Spectral Graph Sparsification: Intuition

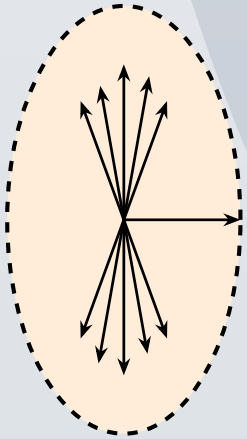
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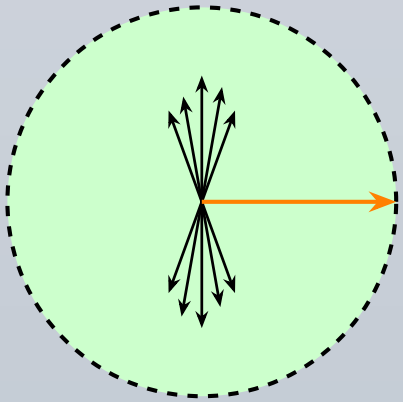
Dumbbell

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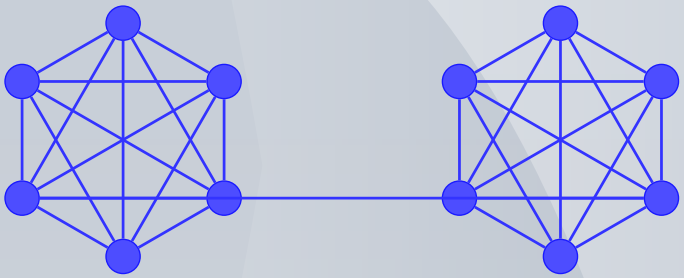
The vector corresponding to the link gets stretched!

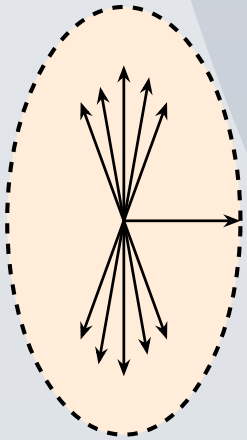
Spectral Graph Sparsification: Intuition

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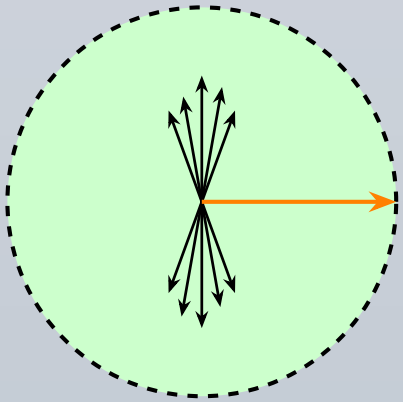
Dumbbell

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$\sum_{e \in E} \mathbf{v}_e \mathbf{v}_e^\top = \mathbf{I}$



The vector corresponding to the link gets stretched!

because this transformation makes all the directions important

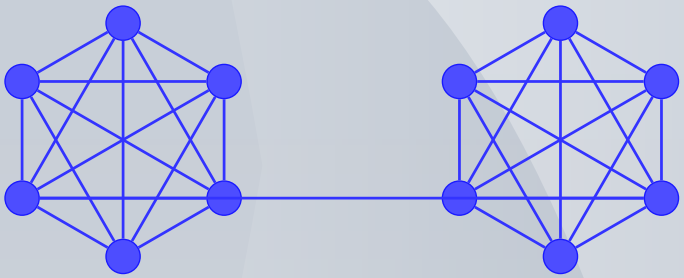
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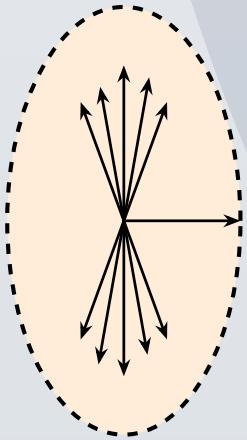
Example: What happens with a dumbbell?

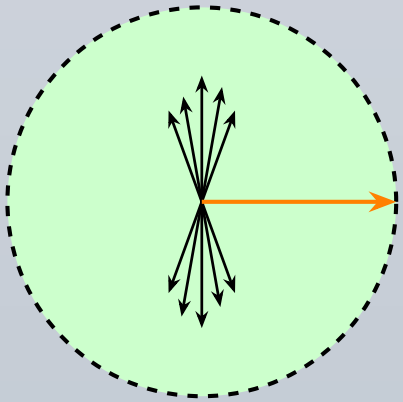
Dumbbell

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The vector corresponding to the link gets stretched!

because this transformation makes all the directions important

rescaling reveals the vectors that are critical

https://math.berkeley.edu/~nikhil/

References I

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Isara Labs



https://misovalko.github.io/mva-ml-graphs.html